Sign and Like-Term Errors in Polynomial Operations Answer Key

Algebra 1 — Expressions and polynomial operations

Quick Answers

1. $2x^2 + 11x - 2$
2. $3x^2 - 10x + 10$
3. $-6x^3 + 12x^2 - 15x$
4. $3x^2 - 4x + 6$, 10
5. $-6x^2 + 19x - 20$, -7
6. $(x - 8)(x + 3)$, -28
7. $4x^2 - 12x + 9$, 1
8. —

Curriculum

Level: Algebra 1 — Expressions and polynomial operations
HSA-APR.A.1 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

4. *If they got -8: distributed the subtraction sign to the first term only, so -2x and -5 kept their original signs*
5. *If they got -35: multiplied -2x by -7 and wrote a negative result, losing the sign change that two negatives produce*
6. *If they got -18: put the minus sign on the smaller factor, writing the pair the other way round*
7. *If they got 13: squared each term on its own and dropped the middle term of the product*

1. Combine like terms: $7x^2 + 3x - 5x^2 + 8x - 2$.

Solution: Group terms with the identical variable part, keeping each coefficient's sign attached to the term in front of it:

$$(7x^2 - 5x^2) + (3x + 8x) - 2$$

The x^2 terms give $2x^2$, the x terms give $11x$, and the constant is unchanged.

$$\boxed{2x^2 + 11x - 2}$$

The x^2 group and the x group are never merged: $2x^2 + 11x$ is already as simple as it gets.

2. Subtract: $(5x^2 - 3x + 4) - (2x^2 + 7x - 6)$.
Solution: Distribute the subtraction sign across all three terms of the second polynomial first — this is the line worth writing every time:

$$5x^2 - 3x + 4 - 2x^2 - 7x + 6$$

Now combine: $5x^2 - 2x^2 = 3x^2$; $-3x - 7x = -10x$; $4 + 6 = 10$. $\boxed{3x^2 - 10x + 10}$
Note the constant grew from 4 to 10. Subtracting -6 adds 6, which is exactly the term students most often leave alone.

3. Multiply: $-3x(2x^2 - 4x + 5)$.
Solution: Multiply each term of the trinomial by $-3x$, tracking both the sign and the exponent:

$$(-3x)(2x^2) = -6x^3, \qquad (-3x)(-4x) = +12x^2, \qquad (-3x)(5) = -15x$$

$\boxed{-6x^3 + 12x^2 - 15x}$
The middle term turns positive because a negative times a negative is positive. A sign pattern that alternates like this is normal, not a warning sign.

4. Find and fix: $(4x^2 - 6x + 1) - (x^2 - 2x - 5)$.
The mistake: the subtraction sign was applied to x^2 only. The student's second and third terms, $-2x$ and -5, were copied down unchanged, so they wrote $-6x - 2x = -8x$ and $1 - 5 = -4$.
Correct work: multiply every term of the second polynomial by -1:

$$4x^2 - 6x + 1 - x^2 + 2x + 5$$

Then $4x^2 - x^2 = 3x^2$; $-6x + 2x = -4x$; $1 + 5 = 6$. $\boxed{3x^2 - 4x + 6}$
Check at $x = 2$: $3(4) - 4(2) + 6 = 12 - 8 + 6$. $\boxed{10}$
The student's expression at $x = 2$ gives -8, so the two are not the same polynomial and the substitution settles it in one line.

5. Find and fix: $-2x(3x - 7) + 5(x - 4)$.
The mistake: the middle term. $(-2x)(-7) = +14x$, not $-14x$; the student carried the 7's minus sign through instead of letting the two negatives cancel. That turned $14x + 5x = 19x$ into $-14x + 5x = -9x$.
Correct work:

$$(-2x)(3x) = -6x^2, \quad (-2x)(-7) = +14x, \quad 5(x) = 5x, \quad 5(-4) = -20$$

so the expansion is $-6x^2 + 14x + 5x - 20$. $\boxed{-6x^2 + 19x - 20}$
Check at $x = 1$: $-6 + 19 - 20$. $\boxed{-7}$
The student's version gives -35 at $x = 1$ — a gap of 28, which is exactly twice the $14x$ term that was mis-signed.

6. Factor $x^2 - 5x - 24$ and check.

Solution: We need two numbers whose product is -24 and whose sum is -5. A negative product means the signs differ, and the sum is negative, so the number with the larger absolute value carries the minus: -8 and $+3$, since $(-8)(3) = -24$ and $-8 + 3 = -5$.

$\boxed{(x-8)(x+3)}$

Check at $x = 1$: the factored form gives $(1-8)(1+3) = (-7)(4)$, and the original gives $1 - 5 - 24$. Both equal -28. $\boxed{-28}$

A single substitution catches a swapped pair immediately: the wrong choice $(x+8)(x-3)$ gives -18 at $x = 1$, not -28.

7. Find and fix: is $(2x-3)^2$ the same as $4x^2 + 9$?

Decision: no. Squaring a binomial is a product of two identical factors, so the cross terms are part of the answer:

$$(2x-3)^2 = (2x-3)(2x-3) = 4x^2 - 6x - 6x + 9$$

$\boxed{4x^2 - 12x + 9}$

Check at $x = 1$: the correct expansion gives $4 - 12 + 9$. $\boxed{1}$

The student's $4x^2 + 9$ gives 13 at $x = 1$. The whole difference is the missing $-12x$, and it never vanishes: only $x = 0$ makes the two expressions agree, which is why testing at $x = 0$ is a poor check.

8. Explain: two claims to correct. *(Open response — flagged for manual review; a model answer follows.)*

Model answer: The first claim is a half-remembered version of the real rule. Subtracting a polynomial means multiplying the entire polynomial by -1 before adding, so every term inside the parentheses changes sign, not just the leading one. Writing the "add the opposite" line explicitly — turning $-(x^2 - 2x - 5)$ into $-x^2 + 2x + 5$ — makes the rule visible instead of remembered.

The second claim confuses adding coefficients with adding terms. $3x^2$ and $5x$ have different variable parts, so they are not like terms and cannot be merged at all. Substituting $x = 2$ proves it: $3(4) + 5(2) = 22$, while $8x^2$ gives $8(4) = 32$. One counterexample is enough to end the argument, and $x = 1$ would not have worked, since both sides give 8 there.

Two terms may be combined only when their variable parts are identical — the same variables raised to the same powers. Coefficients are added; exponents are never touched.

What a reviewer should look for: the subtraction rule stated as multiplying every term by -1, a substitution that actually separates the two expressions, and a definition of like terms that mentions the variable part rather than the coefficients.